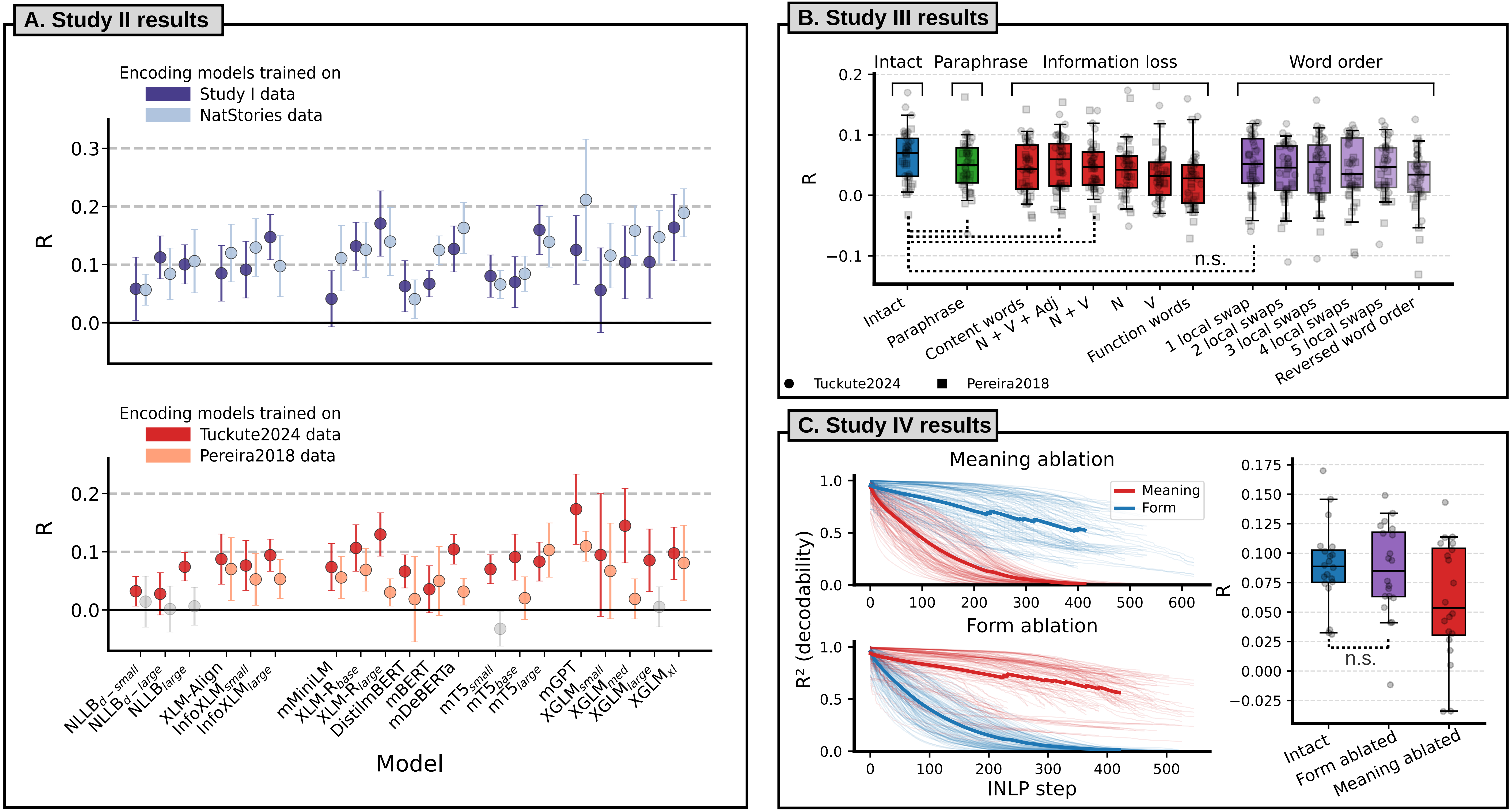
A. Study II results
Encoding models trained on
Study I data
NatStories data
R
0.3
0.2
0.1
0.0
Encoding models trained on
Tuckute2024 data
Pereira2018 data
R
0.2
0.1
0.0
NLLB$_{d-small}$
NLLB$_{d-large}$
NLLB$_{large}$
XLM-Align
InfoXLM$_{small}$
InfoXLM$_{large}$
mMiniLM
XLM-R$_{base}$
XLM-R$_{large}$
DistilmBERT
mBERT
mDeBERTa
mT5$_{small}$
mT5$_{base}$
mT5$_{large}$
mGPT
XGLM$_{small}$
XGLM$_{med}$
XGLM$_{large}$
XGLM$_{xl}$
Model
B. Study III results
Intact
Paraphrase
Information loss
Word order
R
0.2
0.1
0.0
−0.1
n.s.
Intact
Paraphrase
Content words
N + V + Adj
N + V
N
V
Function words
1 local swap
2 local swaps
3 local swaps
4 local swaps
5 local swaps
Reversed word order
● Tuckute2024
■ Pereira2018
C. Study IV results
Meaning ablation
Meaning
Form
1.0
0.5
0.0
0
100
200
300
400
500
600
Form ablation
1.0
0.5
0.0
0
100
200
300
400
500
R² (decodability)
INLP step
R
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
n.s.
Intact
Form ablated
Meaning ablated